Incident Management Automation in ServiceNow

1. Ideation Phase

1.1 Problem Identification

Organizations often receive IT incidents through emails, phone calls, or informal communication. Without a structured Incident Management process, incidents may be:

- Improperly classified
- Assigned to the wrong support team
- Delayed during investigation
- Difficult to track
- Escalated without proper documentation
- Resolved without recording the root cause
- Difficult to use for future knowledge creation

This creates SLA risks and reduces visibility into the overall incident lifecycle.

1.2 Proposed Solution

The proposed solution is to implement a **structured and automated Incident Management lifecycle in ServiceNow**.

The solution will connect:

Incident Management → Knowledge Management → Assignment & Escalation → Change Management → Child Incidents → Resolution → Knowledge Creation → SLA Tracking

1.3 Project Vision

The project aims to provide Service Desk and Level 2 support teams with a standardized process to:

- Create incidents
- Classify and prioritize incidents
- Use knowledge articles for troubleshooting
- Escalate incidents to appropriate teams
- Create emergency changes when required
- Track child incidents

- Document cause and resolution
- Generate knowledge from resolved incidents
- Monitor SLAs and related records

1.4 Project Outcome

The expected outcome is an end-to-end Incident Management process that improves:

- Incident resolution time
- SLA compliance
- Incident classification
- Team collaboration
- Change traceability
- Knowledge reuse
- Service visibility

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2. Requirement Analysis Phase

2.1 Business Objectives

The primary business objectives are:

1. Enable Service Desk agents to create and classify incidents.
2. Provide knowledge-based troubleshooting support.
3. Enable proper reassignment and escalation.
4. Allow Level 2 teams to investigate and resolve incidents.
5. Create emergency Change Requests directly from incidents.
6. Support child incident management.
7. Capture incident cause and resolution details.
8. Generate Knowledge Articles from resolved incidents.
9. Track SLAs and related records.

2.2 Functional Requirements

The system should support:

Incident Management

- Incident creation
- Caller selection

- Short description and description
- Category and subcategory
- Urgency and priority
- Service and service offering
- Configuration Item
- Assignment group
- Assigned to
- Work notes
- Watch list

Knowledge Integration

- Agent Assist
- Knowledge article search
- Article attachment to incident
- Helpful feedback
- Knowledge article creation from resolved incident

Assignment & Escalation

- Service Desk assignment
- Level 2 Network team assignment
- Agent reassignment
- Automatic clearing of Assigned to when assignment group changes

Change Integration

- Create Change Request from incident
- Emergency change selection
- Automatic field population
- Change-to-incident relationship
- Put incident On Hold while awaiting change

Child Incident Management

- Create child incidents
- Link child incidents to parent
- Track child incident status

Resolution

- Capture probable cause
- Capture resolution code
- Capture resolution notes

- Resolve parent incident
- Resolve child incident

SLA Management

- Task SLA visibility
- SLA tracking
- Related records validation

2.3 Stakeholder Requirements

Stakeholder	Role	Requirement
End Users	Report IT issues	Quick resolution and visibility
Service Desk Agents	L1 Support	Easy incident creation and classification
Level 2 Support	Technical Resolution	Complete incident information and ownership
Change Management Team	Change Control	Structured emergency change process
ServiceNow Administrator	Platform Administration	Maintain configuration and automation

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3. Project Design Phase

3.1 Application Design

The solution will use the following ServiceNow modules:

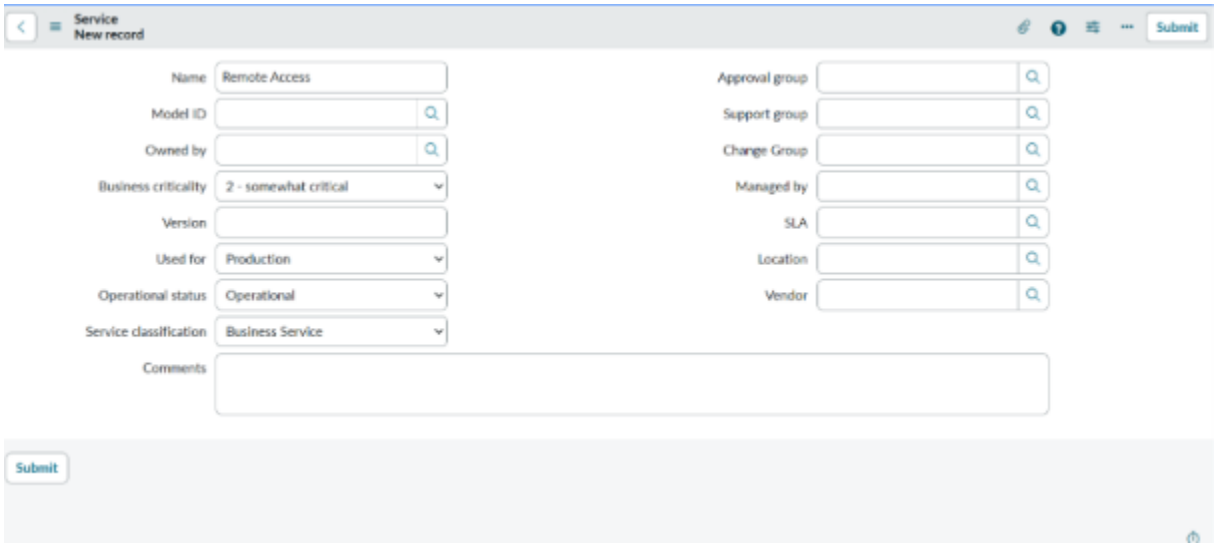
- Incident Management
- Service Configuration
- CMDB
- Knowledge Management
- Change Management
- SLA Management
- Service Operations Workspace

3.2 Data and Configuration Design

The following supporting records will be configured.

Service

Name: Remote Access
Operational Status: Operational



Service Offering

Name: Corporate VPN
Parent: Remote Access
Operational Status: Operational

Configuration Item

Name: VPN Gateway

These records will be used during incident classification.

3.3 Incident Lifecycle Design

The designed lifecycle is:

Incident Creation
↓
Classification

↓
Knowledge Investigation
↓
Service Desk Assignment
↓
Level 2 Escalation
↓
Investigation
↓
Emergency Change if Required
↓
On Hold – Awaiting Change
↓
Resolution
↓
Knowledge Creation
↓
SLA & Related Record Validation

3.4 Integration Design

Incident → Knowledge

Agents use **Agent Assist** to search for relevant knowledge articles.

Incident → Change

When infrastructure changes are required, the agent creates an **Emergency Change Request** directly from the incident.

Incident → Child Incident

Recurring or related user issues can be tracked through child incidents.

Incident → Knowledge

After resolution, the incident can be used as the source for a new Knowledge Article.

Incident → SLA

Task SLAs are associated with the incident to monitor resolution performance.

3.5 User Roles

User	Responsibility
Beth Anglin	Service Desk Agent
David Loo	Level 2 Network Support
Samantha Green	Incident Watch List
ServiceNow Administrator	Configuration and maintenance
Change Management Team	Emergency Change Management

4. Project Planning Phase

4.1 Execution Roadmap

The project will be executed through the following milestones:

Phase	Milestone	Major Activity
Planning	Milestone 1	Service, Service Offering and CI setup
Development	Milestone 2	Incident Record Creation
Development	Milestone 3	Incident Classification
Development	Milestone 4	Knowledge Integration
Development	Milestone 5	Reassignment & Escalation
Development	Milestone 6	Level 2 Investigation
Development	Milestone 7	Emergency Change Creation
Development	Milestone 8	Child Incident Creation
Development	Milestone 9	Incident Resolution

Development	Milestone 10	Knowledge Creation
Validation	Final	SLA and Related Record Validation

4.2 Test Scenario

Business Scenario

A user is unable to connect to the **Corporate VPN** from home.

Incident

Short Description:
Unable to connect to Corporate VPN from home office

Caller

Michael Hoefer

Classification

- Channel: Phone
- Category: Network
- Subcategory: VPN
- Urgency: 2 - Medium
- Service: Remote Access
- Service Offering: Corporate VPN
- Configuration Item: VPN Gateway

Escalation

Incident is initially assigned to **Service Desk** and then reassigned to the **Network** team.

Level 2 Investigation

Network support investigates the issue and identifies that **VPN-SRV-02** requires a restart.

Change

An **Emergency Change Request** is created to restart VPN-SRV-02.

Resolution

The VPN service is restarted and the incident is resolved.

Knowledge

The resolution is converted into a Knowledge Article for future incidents.

4.3 Testing Strategy

Testing will validate:

- Incident creation
- Field classification
- Assignment
- Knowledge search
- Knowledge article attachment
- Reassignment
- Level 2 investigation
- SLA visibility
- Change creation
- Change linkage
- Child incident creation
- Resolution
- Knowledge creation
- Related records

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5. Project Development Phase

This phase contains the actual ServiceNow implementation.

Activity 1: Configure Service

Create:

Remote Access

Path:

All → cmdb_ci_service.LIST → New

Configure:

- Name: Remote Access
- Operational Status: Operational

Activity 2: Configure Service Offering

Create:

Corporate VPN

Path:

All → service_offering.LIST → New

Configure:

- Name: Corporate VPN
- Parent: Remote Access
- Operational Status: Operational

Activity 3: Configure Configuration Item

Create:

VPN Gateway

Path:

All → cmdb_ci.LIST → New

Configure:

- Name: VPN Gateway

Activity 4: Incident Record Creation

Impersonate **Beth Anglin**.

Navigate to:

Service Operations Workspace → List → Incidents → Open → New

Create the incident with:

- Short description: Unable to connect to Corporate VPN from home office
- Caller: Michael Hoefer
- Assignment group: Service Desk

Save the incident.

Activity 5: Incident Classification

Update:

- Description
- Channel: Phone
- Category: Network
- Subcategory: VPN
- Urgency: 2 - Medium
- Service: Remote Access
- Service Offering: Corporate VPN
- Configuration Item: VPN Gateway
- Work notes

Click **Assign to me** and save.

Activity 6: Knowledge Integration

Open **Agent Assist**.

Verify:

- Knowledge Articles are selected
- Relevant articles are displayed
- Article can be opened
- Helpful option works
- Article can be attached

Return to the incident and verify the Knowledge Article appears in the Activity log.

Activity 7: Reassignment & Escalation

Update the Watch List:

- Samantha Green

Add Beth Anglin to the Work notes list.

Change:

Assignment Group → Network

Verify that the **Assigned to** field is cleared automatically.

Save the incident.

Activity 8: Level 2 Investigation

Impersonate **David Loo**.

Open the VPN incident.

Verify:

- Incident description is available
- Knowledge articles are available
- SLA is visible
- Related records are available

Click **Assign to me**.

Activity 9: Emergency Change Request

Update the Configuration Item:

VPN Gateway → VPN-SRV-02

Click:

Create Change Request → Emergency → Continue

Configure:

- Category: Network
- Impact: 1 - High
- Short description: Restart VPN-SRV-02 service
- Description: Multiple users unable to connect. Service appears unresponsive. Restart required.
- Assignment group: Network
- Assigned to: David Loo

Create and save the Change Request.

Return to the incident and set:

- State: On Hold
- On Hold Reason: Awaiting Change

Save.

Activity 10: Child Incident

From **Related Records → Child Incidents**, create a child incident.

Configure:

- Short Description: Unable to connect to Corporate VPN
- Caller: Any user
- Description: User receiving VPN authentication failure error.

Save.

Return to the parent incident and verify that the child incident is displayed.

Activity 11: Incident Resolution

Add the probable cause:

VPN-SRV-02 service was suspended and required restart.

Add resolution:

- Resolution Code: Workaround provided
- Resolution Notes: Restarted VPN-SRV-02 service as per emergency change request.

Save the resolution and click **Resolve**.

Confirm the resolution.

Activity 12: Knowledge Creation

Click:

Create Knowledge

Open:

Related Records → Created Knowledge

Verify that the following information is copied:

- Short Description
- Issue
- Resolution
- Source Task

Publish the Knowledge Article.

Activity 13: Final Validation

Verify the complete lifecycle:

Validation	Expected Result
Parent Incident	Resolved
Child Incident	Resolved
Change Request	Linked to Incident
Knowledge Article	Created

Resolution	Documented
Cause	Documented
SLA	Visible
Related Records	Available
Activity Log	Complete

Final 5-Phase Project Structure

Your complete project can therefore be presented as:

Phase 1 – Ideation Phase

- Problem Identification
- Proposed Solution
- Project Vision
- Expected Business Outcome

Phase 2 – Requirement Analysis

- Business Objectives
- Functional Requirements
- Stakeholder Mapping
- Scope Definition
- Testing Requirements

Phase 3 – Project Design Phase

- Application Design
- Service and CI Design
- Incident Lifecycle Design
- Knowledge Integration Design
- Change Integration Design
- Child Incident Design
- SLA Design
- User/Role Design

Phase 4 – Project Planning Phase

- Execution Roadmap
- Milestones
- Business Scenario
- Test Scenario
- Testing Strategy
- Deployment Validation Plan

Phase 5 – Project Development Phase

1. Service Configuration
2. Service Offering Configuration
3. Configuration Item Creation
4. Incident Creation
5. Incident Classification
6. Knowledge Integration
7. Reassignment & Escalation
8. Level 2 Investigation
9. Emergency Change Creation
10. Child Incident Creation
11. Incident Resolution
12. Knowledge Creation
13. Final Validation